

Binding-Constraint Migration in Advanced-Node Semiconductor Supply: A Vintage-Structured Model with a Diagnostic Temporal Evaluation

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Abstract—This paper develops a continuous-time vintage-structured model of advanced-node semiconductor supply and evaluates it on a later calendar partition using leakage-free target construction. Because the 2013–2025 period was inspected during prior model development, the exercise is a diagnostic temporal evaluation rather than an independent prospective test. Eighteen technology-node vintages carry separate capacity and yield-learning states. Fabrication, equipment, and advanced packaging are linked by a differentiable serial soft minimum, $Q = (\sum_i Q_i^{-\rho})^{-1/\rho}$, which cannot exceed the tightest stage and converges to the hard minimum as $\rho \rightarrow \infty$. The empirical target is U.S. NAICS 3344 capacity utilization constructed from Federal Reserve production and capacity series. Five-year structural smoothing is computed separately within the 1990–2012 estimation window and the 2013–2025 evaluation window. The selected four-parameter S1 fit has training RMSE 0.0967 and later-period RMSE 0.2710, versus 0.0503 for the training-mean benchmark, 0.0813 for a training trend, and 0.0481 for persistence. Nested 6-, 8-, and 10-parameter specifications selected by training error also fail to beat the constant. A scaled-Jacobian diagnostic has condition number 289 and the fitted sharpness lies near the upper search boundary. Under the corrected formulation, the selected S1 policy bundles change the 2042 output index by less than 0.001 percent relative to baseline; the exact near-zero magnitude is not interpreted as a precision effect estimate. A local elasticity decomposition provides an economic interpretation of constraint migration: the modeled marginal leverage of additional capability moves among fabrication, advanced packaging, and demand over time. These elasticities are conditional model quantities, not observed investment returns or monetary valuations. Constraint migration and joint tightness remain model features, with the two tightest stages within 10 percent for 48.7 percent of 2023–2035. The defensible empirical conclusion is limited: predictive skill has not been demonstrated on this target. Because the later period is not independent of prior development, its RMSE should not be interpreted as an unbiased estimate of prospective forecast error.

Index Terms—Semiconductor supply chain, system dynamics, temporal evaluation, model validation, advanced packaging, capacity utilization, bottleneck modeling, capital allocation.

I. INTRODUCTION

THE policy relevance of semiconductor capacity models depends on whether their binding constraints are empirical results or consequences of model construction. If fabrication is limiting, additional fab capacity can increase usable output; if packaging, equipment, workforce, or demand

is tighter, the same intervention can have little effect. Supply-chain models should therefore distinguish arithmetic consequences of a bottleneck operator from conclusions supported by external data [3], [4], [6], [7].

This study uses a vintage-structured system-dynamics model to examine long-horizon migration among fabrication, equipment, and advanced-packaging constraints. The original implementation had two material problems. First, a five-year centered smoother was calculated before the training/evaluation split, allowing 2013–2014 observations into the final training targets. Second, a normalized negative-order generalized mean could exceed the smallest serial-stage capability, so the reported activity index was not physically bounded by the tightest stage or by demand. Both are corrected here.

The resulting paper is intentionally diagnostic. It reports failure against naive benchmarks on the separated later-period partition, optimizer instability, practical-identifiability diagnostics, policy results that change materially after the throughput correction, and an attempted global-sensitivity calculation that fails its predeclared convergence criterion. The later-period result can establish that predictive skill has not been demonstrated by the final model, but it cannot provide an unbiased estimate of prospective generalization error because those calendar years were inspected during earlier development.

Contributions. The rebuilt analysis provides: (1) node-vintage capacity and learning dynamics; (2) a differentiable serial soft minimum that obeys $Q \leq \min_i Q_i$; (3) leakage-free partition-local target smoothing; (4) multi-start nested model selection against naive benchmarks; (5) local practical-identifiability diagnostics and a sharpness profile; (6) policy and bottleneck scenarios regenerated from the corrected model; (7) a local output-elasticity decomposition that connects constraint migration to model-implied marginal system leverage without imposing unsupported monetary assumptions; and (8) a self-contained replication package in which manuscript tables and figures are generated from the same corrected outputs.

II. DATA

Two Federal Reserve G.17 series for NAICS 3344 are used: industrial production (IPG3344S) and industrial capacity (CAPG3344S). Their ratio defines implied capacity utilization,

$$u(t) = \frac{\text{IPG3344S}(t)}{\text{CAPG3344S}(t)}. \quad (1)$$

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TABLE I
OBSERVED STRUCTURAL REGIMES. CAPACITY GROWTH IS ENDPOINT
CAGR OF ANNUAL-MEAN CAPG3344S WITHIN EACH ERA,
RECOMPUTED FROM THE PACKAGED DATA.

Era	Mean utilization	Capacity growth
1990–1999	0.830	35.0%/yr
2000–2009	0.760	21.7%/yr
2010–2019	0.746	11.0%/yr
2020–2025	0.822	8.8%/yr

The Federal Reserve defines industry utilization as output divided by the corresponding capacity index, where capacity represents sustainable maximum output under realistic operating conditions [46], [47]. The ratio is dimensionless and independent of the model inputs.

Annual utilization is aggregated from the packaged monthly series. Because the model omits business-cycle dynamics, estimation uses a five-year centered structural target. Crucially, smoothing is performed *separately* within 1990–2012 and 2013–2025. No observation from the evaluation partition enters an estimation target. A trailing five-year mean is retained as a robustness target.

The target remains imperfect. NAICS 3344 covers U.S.-located semiconductor and other electronic-component production, while the model represents global advanced-node logic. Failure on this target can therefore reflect model error, population mismatch, or both.

III. MODEL

A. Node vintages

Each of 18 technology-node vintages carries installed capacity K_v and cumulative experience E_v . Yield follows

$$Y_v = Y_{\min} + (Y_{\max} - Y_{\min}) \left(1 - e^{-\beta E_v / E_{\text{ref}}}\right), \quad (2)$$

so each new node introduces immature capacity rather than inheriting a single monotone industry-wide yield curve [20], [21]. Completed fab capacity is allocated preferentially to the leading edge; vintages retire at rate δ_{node} .

B. Feasible serial bottleneck

Let Q_{fab} , Q_{equip} , and Q_{pkg} denote positive stage capabilities. The corrected supply operator is

$$Q_{\text{supply}} = \left(Q_{\text{fab}}^{-\rho} + Q_{\text{equip}}^{-\rho} + Q_{\text{pkg}}^{-\rho}\right)^{-1/\rho}. \quad (3)$$

For $q_{\min} = \min_i Q_i$, each ratio $Q_i/q_{\min} \geq 1$, hence the sum of inverse powers is at least one and $Q_{\text{supply}} \leq q_{\min}$. The operator is smooth for positive inputs and converges to the hard minimum as $\rho \rightarrow \infty$. A non-tight stage can therefore have a small nonzero marginal effect without permitting serial throughput above the tightest stage.

Demand is imposed through the same feasible two-input operator,

$$Q_{\text{eff}} = \left(Q_{\text{supply}}^{-\rho} + D(t)^{-\rho}\right)^{-1/\rho}, \quad (4)$$

so $Q_{\text{eff}} \leq \min(Q_{\text{supply}}, D)$. Numerical audit over the full baseline trajectory confirms maximum ratios

TABLE II
LEAKAGE-CORRECTED TEMPORAL-EVALUATION PERFORMANCE. LOWER
RMSE IS BETTER.

	RMSE	Skill vs. constant
S1, training (1990–2012)	0.0967	—
S1, evaluation (2013–2025)	0.2710	−4.39
Training-mean benchmark	0.0503	0
Linear-trend benchmark	0.0813	−0.62
Persistence benchmark	0.0481	+0.04 relative to mean*

*The persistence RMSE is reported directly; the row is not a skill score against itself.

$$Q_{\text{supply}} / \min(Q_c) = 1.0000, \quad Q_{\text{eff}} / Q_{\text{supply}} \leq 1, \quad \text{and} \quad Q_{\text{eff}} / D \leq 1.$$

Stage slack is reported relative to the tightest physical stage,

$$\sigma_c(t) = \frac{Q_c(t)}{\min_{c'} Q_{c'}(t)} - 1. \quad (5)$$

The model has 47 active states: 36 vintage capacity/experience states, packaging and workforce stocks, and nine Erlang pipeline states. The previously inert perceived-utilization state and its smoothing parameter have been removed. Production runs use fourth-order Runge–Kutta with $\Delta t = 1/16$ yr; the annual-utilization difference relative to $\Delta t = 1/4$ has maximum 0.00237 and RMSE 0.00101 at the selected S1 vector.

IV. ESTIMATION AND TEMPORAL EVALUATION

A. Protocol

S1 estimates four parameters: target utilization, packaging throughput, soft-min sharpness ρ , and secular demand growth. Parameters are selected using 1990–2012 training error only under wide bounds and multiple differential-evolution starts. The 2013–2025 partition is excluded from numerical fitting and model selection. Because this research program previously inspected the same calendar years during model development, the comparison is a *diagnostic temporal evaluation*, not an independent prospective holdout. It is used only to test whether predictive skill has been demonstrated by the final specification; it is not used as an unbiased estimate of future forecast accuracy.

Three benchmarks are fitted using training data only: the training mean, a linear trend, and persistence [30], [31]. The selected S1 parameters are $U^* = 0.9744$, $a_{\text{pkg}} = 2.1552$, $\rho = 79.47$, and $g_D = 0.09571$.

The S1 model loses decisively. Against a causal trailing-five-year target, with no refit, its evaluation RMSE is 0.2591 versus 0.0540 for the training-mean benchmark and 0.0469 for persistence. The poor later-period performance therefore does not depend on the centered evaluation smoother.

B. Nested model selection and optimizer instability

Increasing flexibility improves training fit but does not yield a training-selected specification that beats the constant benchmark.

Optimizer-start spread is itself material. For S3, seed 1 gives training RMSE 0.0650 and evaluation RMSE 0.0454,

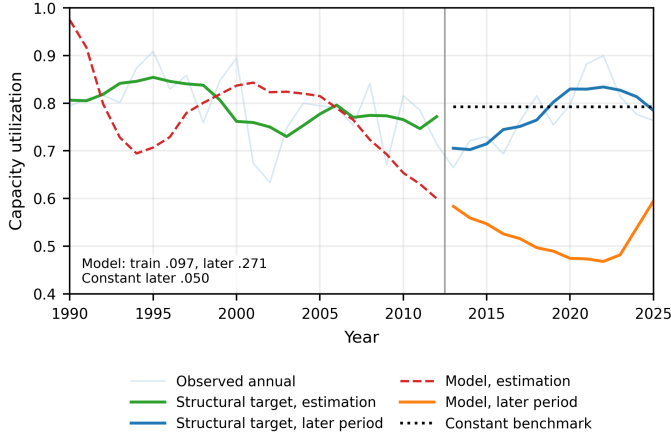


Fig. 1. Leakage-corrected estimation and diagnostic later-period evaluation against Federal Reserve utilization.

TABLE III
NESTED SPECIFICATIONS SELECTED BY LOWEST TRAINING RMSE
ACROSS OPTIMIZER STARTS.

Spec	k	Train RMSE	Eval. RMSE	Skill
S1	4	0.0967	0.2710	-4.39
S2	6	0.0714	0.1128	-1.24
S3	8	0.0638	0.0770	-0.53
S4	10	0.0467	0.0758	-0.51
Constant	0	—	0.0503	0

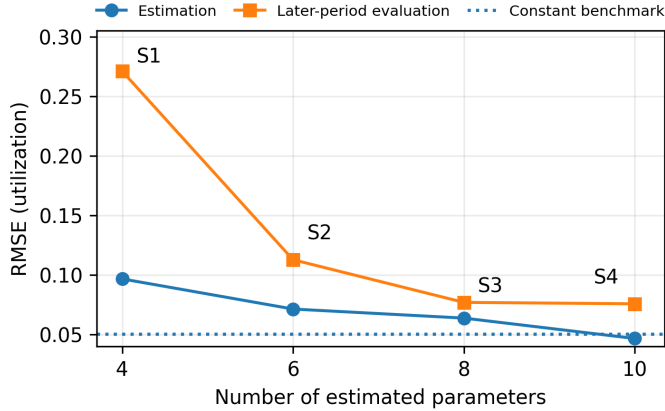


Fig. 2. Training fit improves with flexibility; training-selected evaluation performance remains worse than the constant benchmark.

while seed 2 gives the lower training RMSE 0.0638 but evaluation RMSE 0.0770. Selecting the favorable evaluation run would therefore use evaluation performance for selection and constitute leakage. The lower-training-error seed is retained. S2 likewise shows large evaluation spread across starts. This instability weakens structural interpretation even when an individual run happens to score well.

C. Practical identifiability

A finite-difference Jacobian of the 23 training predictions with respect to the four S1 parameters is scaled by each parameter's search-range width. Its singular values are 3.343, 1.617,

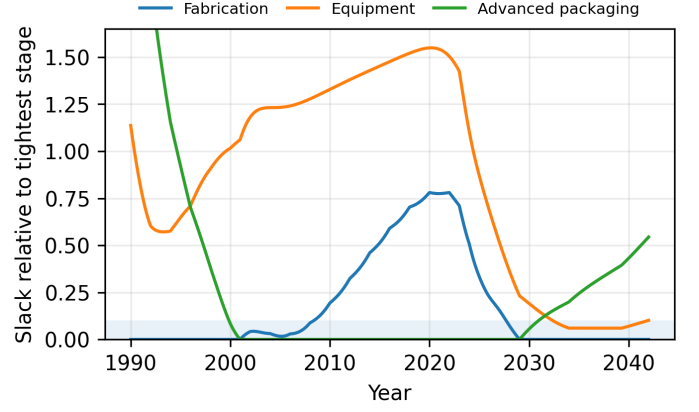


Fig. 3. Stage slack in the corrected baseline. The shaded region marks stages within 10 percent of the tightest stage.

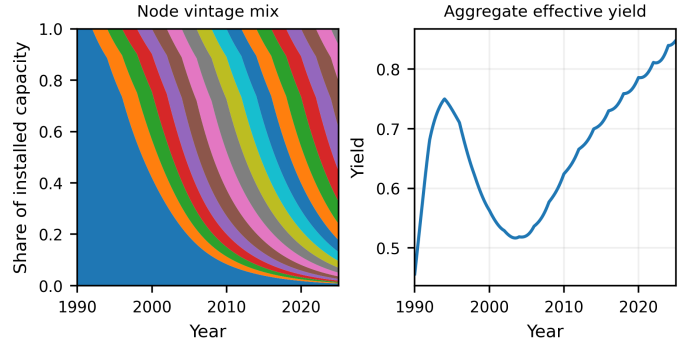


Fig. 4. Node-vintage capacity mix and aggregate effective yield.

0.331, and 0.0116, giving condition number 289. The local inverse-information approximation gives correlation -0.797 between U^* and ρ . A sharpness profile holding the other selected parameters fixed reduces training RMSE from 0.140 at $\rho = 10$ to 0.09669 at $\rho = 80$, 0.09650 at $\rho = 200$, and 0.09651 at $\rho = 500$. The objective therefore becomes nearly flat toward the hard-minimum limit. Finite ρ is weakly identified, and the selected value should not be interpreted as a physical elasticity.

V. STRUCTURAL SCENARIOS

A. Constraint migration and joint tightness

Under selected S1 parameters, fabrication is the tightest stage through 2000, packaging becomes tightest around 2001, and fabrication becomes tightest again around 2029. Over 2023–2035 the two tightest stages are within 10 percent of one another for 48.7 percent of simulated time. Continuous slack therefore remains useful even though the diagnostic later-period comparison is poor.

B. Vintage structure

The vintage formulation produces non-monotone aggregate effective yield because each node introduction injects immature capacity before learning accumulates.

TABLE IV
EFFECTIVE OUTPUT INDEX, COMMON BASELINE 2022 = 100.
CONDITIONAL MODEL SCENARIOS, NOT FORECASTS.

Scenario	2027	2032	2042	20-yr gain
Baseline	234.2	380.1	990.0	—
Capital-only	234.2	380.1	990.0	< 0.001%
Capital + workforce	234.2	380.1	990.0	< 0.001%
Ecosystem-aligned	234.2	380.1	990.0	< 0.001%

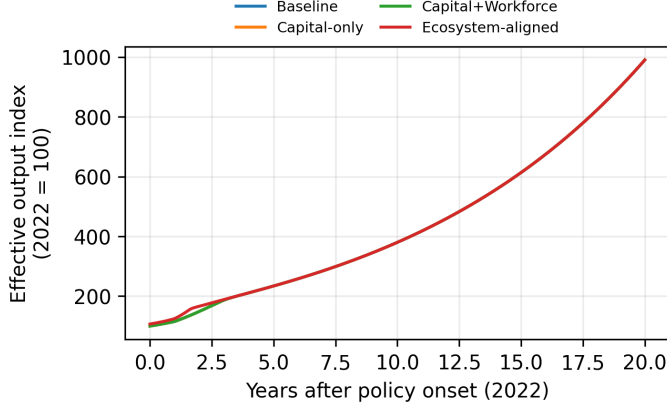


Fig. 5. Policy bundles regenerated under the feasible serial soft minimum.

C. Policy scenarios

The physically feasible bottleneck reverses the earlier policy result. All four scenarios have the same reported index to one decimal place.

This is not evidence that real supply-side policy is ineffective. It is evidence that the previous +23.2 percent result was operator-dependent. At the selected S1 vector, the exact 2042 gain is approximately 3×10^{-7} percent, but that number is an asymptotic numerical consequence of the near-hard-minimum setting and is not treated as a precision effect estimate. Holding the other selected S1 parameters fixed, a one-dimensional check over $\rho = 10, 20, 40, 79.47$ gives a largest 2042 gain across the three policy bundles of approximately 0.13, 0.026, 0.00064, and 3×10^{-7} percent, respectively. The defensible statement is therefore that the modeled long-run policy gain is small across this sharpness check and negligible at the selected vector. Exogenous demand becomes tighter than all supply stages from approximately 2025 onward, further suppressing long-horizon differences.

D. Economic interpretation: marginal system leverage

The serial soft-min permits a direct local economic interpretation without adding unsupported price, cost, or revenue assumptions. Because the nested supply-and-demand operator is algebraically equivalent to

$$Q_{\text{eff}} = \left(\sum_{i \in \{\text{fab}, \text{equip}, \text{pkg}, D\}} Q_i^{-\rho} \right)^{-1/\rho}, \quad (6)$$

the local output elasticity with respect to capability Q_i is

$$\eta_i(t) = \frac{\partial \ln Q_{\text{eff}}}{\partial \ln Q_i} = \left(\frac{Q_{\text{eff}}}{Q_i} \right)^\rho. \quad (7)$$

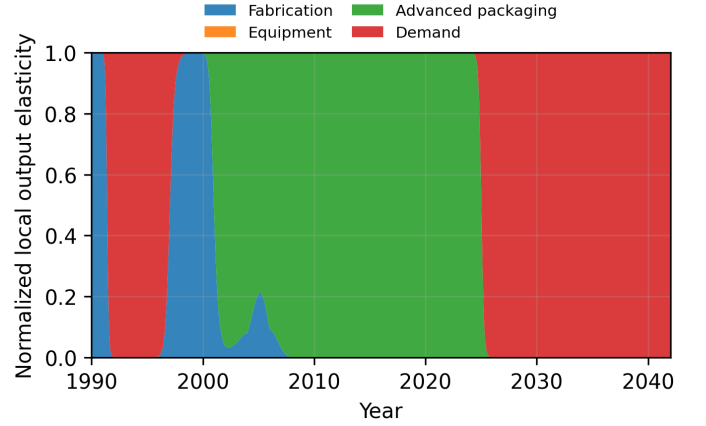


Fig. 6. Model-implied marginal system leverage. The stacked shares are the local output elasticities in (7) and sum to one at each time. They are conditional structural quantities, not observed investment returns, prices, or monetary valuations.

The operator is homogeneous of degree one, so $\sum_i \eta_i = 1$. The η_i can therefore be read as normalized shares of local system leverage: a stage with η_i near one has nearly all of the modeled marginal percentage response to an incremental percentage increase in capability, whereas a stage with η_i near zero is locally slack relative to another constraint.

Figure 6 shows the decomposition for the selected S1 trajectory. Fabrication carries most modeled marginal leverage in the early 1990s and again around the late 1990s; advanced packaging dominates from approximately 2001 through 2025; demand dominates after approximately 2025. Equipment does not become the dominant marginal constraint in this trajectory. The economic implication is conditional rather than monetary: as the active constraint migrates, the modeled marginal system leverage of adding capacity migrates with it. Investment in a slack stage can therefore have little immediate effect on effective output until complementary constraints are relieved.

A monetary ranking would require credible stage-specific marginal costs. If such costs $c_i(t)$ were available, a cost-adjusted extension could compare η_i/c_i , but the present paper does not estimate that quantity. This avoids converting a structural model into an unsupported NPV or return-on-investment calculation.

E. Sensitivity diagnostic

A seven-input Jansen variance-decomposition exercise [25], [22], [27] was attempted using a scrambled Sobol design with base sizes $N = 8, 16, 32$. The calculation failed the predeclared $\max |\Delta S_T| < 0.05$ convergence rule for both analyzed outputs (0.182 for the 20-year output index and 0.142 for the packaging-tightest fraction). The resulting indices and rankings are therefore *not reported as quantitative findings*. The failed diagnostic is retained in the replication materials to make clear that a substantially larger and demonstrably converged design is required before variance-based sensitivity claims are made.

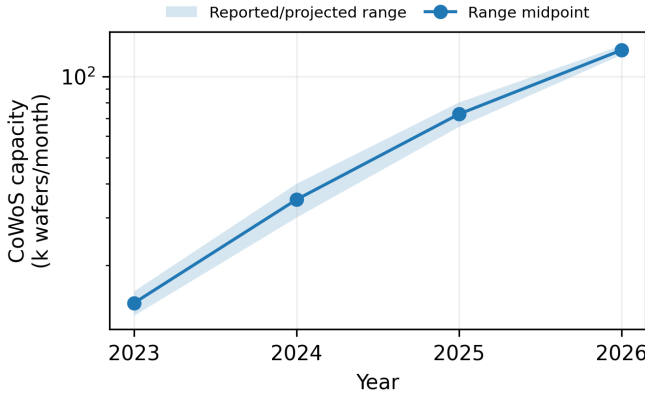


Fig. 7. Reported, estimated, and projected CoWoS capacity anchors. The 2026 point is a projection.

VI. PACKAGING CONTEXT

The packaged external-anchor file compiles TSMC CoWoS capacity ranges for 2023–2026. These points are heterogeneous third-party reported estimates and projections, not a homogeneous audited historical series. The later values are therefore labeled reported/estimated or projected in the corrected package. Their midpoint trajectory corresponds to approximately 105 percent annualized growth, but that number is descriptive of the compiled anchors and is not used as a validation statistic. Extended packaging lead-time reports and heterogeneous-integration roadmaps provide qualitative context only [43].

VII. LIMITATIONS

First, the 2013–2025 evaluation is leakage-free numerically but not independent of prior model development because those years were inspected earlier. It therefore supports only the limited conclusion that predictive skill has not been demonstrated by the final model; it does not provide an unbiased estimate of prospective forecast error. A confirmatory test requires newly released data or an independent advanced-node-specific target.

Second, the empirical target and modeled population do not match exactly. NAICS 3344 is U.S.-located and broader than global advanced-node logic.

Third, the fitted sharpness is practically weakly identified and trends toward the hard-minimum limit. Structural conclusions that depend on finite substitution should therefore be treated cautiously.

Fourth, many fixed coefficients and policy strengths are modeling assumptions rather than independently estimated quantities. The attempted restricted sensitivity calculation failed its convergence criterion at $N = 32$; no numerical sensitivity ranking is treated as a result.

Fifth, demand is exogenous and the model lacks prices, investment costs, revenues, inventories, geography, trade flows, supplier-level heterogeneity, and explicit product mix. The elasticity decomposition is therefore a model-implied marginal-throughput diagnostic, not an observed financial return. These omissions limit causal and monetary policy interpretation.

VIII. CONCLUSION

After correcting target leakage and enforcing physically feasible serial throughput, the final model still performs worse than naive benchmarks on the diagnostic later-period partition, and the earlier long-run supply-policy gain disappears. This changes the scientific emphasis of the study. The strongest result is not a forecast or a policy prescription; it is a reproducible demonstration that semiconductor bottleneck conclusions are highly sensitive to structural assumptions and that apparent model skill can be unstable across parameter bounds and optimizer starts. The local elasticity decomposition adds a decision-oriented interpretation: within the model, marginal system leverage migrates with the active constraint, so capacity added to a slack stage can have little immediate throughput value. This is an economic interpretation of the structural model, not a monetary valuation.

The vintage formulation and continuous slack diagnostics remain useful for structural experimentation. Under the selected S1 trajectory, packaging is tightest for much of the 2000s and 2010s, the binding stage later migrates back toward fabrication, and joint tightness is common in 2023–2035. But these scenarios are conditional on a model that does not predict the aggregate utilization target well. The appropriate next empirical advance is a new external target closer to global advanced-node production, combined with a genuinely prospective evaluation period.

APPENDIX A COMPLETE PARAMETER SET REPLICATION

The research replication materials contain the source data, rebuilt model, leakage-free target construction, calibration records, nested model-selection results, identifiability diagnostics, feasibility checks, policy calculations, the marginal-elasticity decomposition, the failed sensitivity-convergence diagnostic, figure scripts, and this manuscript source.

REFERENCES

- [1] J. W. Forrester, *Industrial Dynamics*. Cambridge, MA: MIT Press, 1961.
- [2] J. D. Sterman, *Business Dynamics: Systems Thinking and Modeling for a Complex World*. Boston, MA: Irwin/McGraw-Hill, 2000.
- [3] Y. Barlas, “Formal aspects of model validity and validation in system dynamics,” *System Dynamics Review*, vol. 12, no. 3, pp. 183–210, 1996.
- [4] J. W. Forrester and P. M. Senge, “Tests for building confidence in system dynamics models,” *TIMS Studies in the Management Sciences*, vol. 14, pp. 209–228, 1980.
- [5] J. D. W. Morecroft, *Strategic Modelling and Business Dynamics*. Chichester, UK: Wiley, 2007.
- [6] N. Oreskes, K. Shrader-Frechette, and K. Belitz, “Verification, validation, and confirmation of numerical models in the earth sciences,” *Science*, vol. 263, no. 5147, pp. 641–646, 1994.
- [7] J. B. Homer, “Why we iterate: scientific modeling in theory and practice,” *System Dynamics Review*, vol. 12, no. 1, pp. 1–19, 1996.
- [8] L. Mönch, R. Uzsoy, and J. W. Fowler, “A survey of semiconductor supply chain models Part I,” *Int. J. Production Research*, vol. 56, no. 13, pp. 4524–4545, 2018.
- [9] L. Mönch, R. Uzsoy, and J. W. Fowler, “A survey of semiconductor supply chain models Part II,” *Int. J. Production Research*, vol. 56, no. 13, pp. 4546–4564, 2018.
- [10] L. Mönch, R. Uzsoy, and J. W. Fowler, “A survey of semiconductor supply chain models Part III,” *Int. J. Production Research*, vol. 56, no. 13, pp. 4565–4584, 2018.

TABLE V

COMPLETE PARAMETER SET FOR THE SELECTED S1 REBUILD.
PARAMETERS MARKED *calibrated* ARE ESTIMATED ON 1990–2012; ALL OTHERS ARE FIXED MODELING ASSUMPTIONS. INITIAL CONDITIONS ARE LISTED AT THE END.

Symbol	Description	Value	Source
t0	Start year	1990	fixed
t_end	Estimation horizon	2025	fixed
t_horizon	Projection horizon	2042	fixed
dt	Integration step	0.0625	fixed
delta_node	Vintage capacity retirement rate	0.05	fixed
leading_edge_share	Share of new capacity to newest node	0.4	fixed
y_min	Yield at node introduction	0.2	fixed
y_max	Mature asymptotic yield	0.92	fixed
beta	Learning speed per unit normalized experience	0.55	fixed
E_ref	Experience normalizer	1	fixed
delta_pkg	Packaging depreciation	0.07	fixed
tau_pkg_base	Packaging build delay, pre-2023	2.5	fixed
tau_pkg_fast	Packaging build delay, post-2023	1.75	fixed
pkg_year_break	Packaging delay break year	2023	fixed
n_pkg	Packaging Erlang order	3	fixed
a_pkg	Packaging throughput per unit capacity	2.155	calibrated
tau_fab	Fabrication build+qualification delay	4	fixed
n_fab	Fabrication Erlang order	3	fixed
tau_w	Workforce formation delay	6	fixed
n_w	Workforce Erlang order	3	fixed
delta_w	Workforce attrition	0.04	fixed
w_target_ratio	Target workforce-to-capacity ratio	1	fixed
a_eq	Tool availability fraction	0.95	fixed
rho	Soft-min sharpness	79.47	calibrated
demand_0	Initial demand level	60	fixed
demand_growth	Secular demand growth	0.09571	calibrated
ai_surge_year	AI surge inflection	2022	fixed
ai_surge_width	AI surge width	1.2	fixed
ai_surge_amp	AI packaging-demand surge amplitude	1.1	fixed
ai_demand_amp	AI total-demand lift	0.35	fixed
inv_gain	Investment responsiveness to demand gap	0.55	fixed
g_base	Autonomous fab investment rate	0.16	fixed
base_invest	Autonomous investment floor	6	fixed
util_target	Target utilization	0.9744	calibrated
pkg_inv_gain	Packaging investment responsiveness	0.5	fixed
g_pkg_base	Autonomous packaging investment rate	0.14	fixed
pkg_base_invest	pkg base invest	2	fixed
hire_gain	Hiring responsiveness	0.5	fixed
p_capital	p capital	0	fixed
p_workforce	p workforce	0	fixed
p_ecosystem	p ecosystem	0	fixed
policy_start	Policy onset year	2022	fixed
cap_strength	Capital lever strength	1.6	fixed
wf_strength	Workforce lever strength	1	fixed
eco_pkg_strength	Ecosystem lever, packaging starts	1.2	fixed
eco_pkg_throughput	Ecosystem lever, packaging throughput	0.15	fixed
eco_equip_strength	Ecosystem lever, tool availability	0.03	fixed
K_total0	Initial total fab capacity	100	fixed
K_pkg0	Initial packaging capacity	95	fixed
W0	Initial workforce	100	fixed
E_seed	Experience seed, pre-1990 vintages	0.8	fixed

- [11] J. Lin, V. L. M. Spiegler, and M. M. Naim, “Dynamic analysis and design of a semiconductor supply chain: a control engineering approach,” *Int. J. Production Research*, vol. 56, no. 13, pp. 4585–4611, 2018.
- [12] V. L. M. Spiegler, M. M. Naim, and J. Wikner, “A control engineering approach to the assessment of supply chain resilience,” *Int. J. Production Research*, vol. 50, no. 21, pp. 6162–6187, 2012.
- [13] H. L. Lee, V. Padmanabhan, and S. Whang, “Information distortion in a supply chain: the bullwhip effect,” *Management Science*, vol. 43, no.

- 4, pp. 546–558, 1997.
- [14] P. Gonçalves, J. Hines, and J. Sterman, “The impact of endogenous demand on push–pull production systems,” *System Dynamics Review*, vol. 21, no. 3, pp. 187–216, 2005.
- [15] T. P. Wright, “Factors affecting the cost of airplanes,” *Journal of the Aeronautical Sciences*, vol. 3, no. 4, pp. 122–128, 1936.
- [16] L. Argote and D. Epple, “Learning curves in manufacturing,” *Science*, vol. 247, no. 4945, pp. 920–924, 1990.
- [17] D. A. Irwin and P. J. Klenow, “Learning-by-doing spillovers in the semiconductor industry,” *Journal of Political Economy*, vol. 102, no. 6, pp. 1200–1227, 1994.
- [18] H. Gruber, *Learning and Strategic Product Innovation: Theory and Evidence for the Semiconductor Industry*. Amsterdam: North-Holland, 1994.
- [19] N. W. Hatch and D. C. Mowery, “Process innovation and learning by doing in semiconductor manufacturing,” *Management Science*, vol. 44, no. 11, pp. 1461–1477, 1998.
- [20] C. Weber, “Yield learning and the sources of profitability in semiconductor manufacturing and process development,” *IEEE Trans. Semiconductor Manufacturing*, vol. 17, no. 4, pp. 590–596, 2004.
- [21] I. Tirkel, “Yield learning curve models in semiconductor manufacturing,” *IEEE Trans. Semiconductor Manufacturing*, vol. 26, no. 4, pp. 564–571, 2013.
- [22] I. M. Sobol’, “Global sensitivity indices for nonlinear mathematical models and their Monte Carlo estimates,” *Mathematics and Computers in Simulation*, vol. 55, no. 1–3, pp. 271–280, 2001.
- [23] A. Saltelli et al., *Global Sensitivity Analysis: The Primer*. Chichester, UK: Wiley, 2008.
- [24] A. Saltelli, P. Annoni, I. Azzini, F. Campolongo, M. Ratto, and S. Tarantola, “Variance based sensitivity analysis of model output: design and estimator for the total sensitivity index,” *Computer Physics Communications*, vol. 181, no. 2, pp. 259–270, 2010.
- [25] M. J. W. Jansen, “Analysis of variance designs for model output,” *Computer Physics Communications*, vol. 117, no. 1, pp. 35–43, 1999.
- [26] J. Herman and W. Usher, “SALib: an open-source Python library for sensitivity analysis,” *Journal of Open Source Software*, vol. 2, no. 9, p. 97, 2017.
- [27] B. Iooss and P. Lemaître, “A review on global sensitivity analysis methods,” in *Uncertainty Management in Simulation-Optimization of Complex Systems*. Boston, MA: Springer, 2015, pp. 101–122.
- [28] K. P. Burnham and D. R. Anderson, *Model Selection and Multimodel Inference*, 2nd ed. New York: Springer, 2002.
- [29] S. Arlot and A. Celisse, “A survey of cross-validation procedures for model selection,” *Statistics Surveys*, vol. 4, pp. 40–79, 2010.
- [30] R. J. Hyndman and G. Athanasopoulos, *Forecasting: Principles and Practice*, 3rd ed. Melbourne, Australia: OTexts, 2021.
- [31] F. X. Diebold and R. S. Mariano, “Comparing predictive accuracy,” *Journal of Business & Economic Statistics*, vol. 13, no. 3, pp. 253–263, 1995.
- [32] M. A. Babyak, “What you see may not be what you get: a brief, nontechnical introduction to overfitting in regression-type models,” *Psychosomatic Medicine*, vol. 66, no. 3, pp. 411–421, 2004.
- [33] K. J. Arrow, H. B. Chenery, B. S. Minhas, and R. M. Solow, “Capital-labor substitution and economic efficiency,” *Review of Economics and Statistics*, vol. 43, no. 3, pp. 225–250, 1961.
- [34] L. Johansen, “Substitution versus fixed production coefficients in the theory of economic growth: a synthesis,” *Econometrica*, vol. 27, no. 2, pp. 157–176, 1959.
- [35] R. M. Solow, “Investment and technical progress,” in *Mathematical Methods in the Social Sciences*, K. J. Arrow, S. Karlin, and P. Suppes, Eds. Stanford, CA: Stanford Univ. Press, 1960, pp. 89–104.
- [36] J. T. Macher, D. C. Mowery, and D. A. Hodges, “Reversal of fortune? The recovery of the U.S. semiconductor industry,” *California Management Review*, vol. 41, no. 1, pp. 107–136, 1998.
- [37] K. Flamm, *Mismanaged Trade? Strategic Policy and the Semiconductor Industry*. Washington, DC: Brookings Institution Press, 1996.
- [38] W. B. Arthur, “Competing technologies, increasing returns, and lock-in by historical events,” *Economic Journal*, vol. 99, no. 394, pp. 116–131, 1989.
- [39] A. Varas, R. Varadarajan, J. Goodrich, and F. Yinug, “Strengthening the global semiconductor supply chain in an uncertain era,” Boston Consulting Group and Semiconductor Industry Association, Apr. 2021.
- [40] S. M. Khan, A. Mann, and D. Peterson, “The semiconductor supply chain: assessing national competitiveness,” Center for Security and Emerging Technology, Jan. 2021.
- [41] Congressional Research Service, “Semiconductors and the CHIPS Act: the global context,” CRS Report, 2023.

- [42] Semiconductor Industry Association, *State of the U.S. Semiconductor Industry*, 2025.
- [43] IEEE Electronics Packaging Society, *Heterogeneous Integration Roadmap*, 2023.
- [44] C. Corrado and J. Matthey, “Capacity utilization,” *Journal of Economic Perspectives*, vol. 11, no. 1, pp. 151–167, 1997.
- [45] Board of Governors of the Federal Reserve System (US), “Industrial production and capacity utilization—G.17,” statistical release.
- [46] Board of Governors of the Federal Reserve System (US), “Industrial Production: Semiconductor and Other Electronic Component (NAICS 3344) [IPG3344S],” retrieved from FRED, Federal Reserve Bank of St. Louis.
- [47] Board of Governors of the Federal Reserve System (US), “Industrial Capacity: Semiconductor and Other Electronic Component (NAICS 3344) [CAPG3344S],” retrieved from FRED, Federal Reserve Bank of St. Louis.